

145. $9H + H8 + H6 = 230 \Rightarrow \{(9 \times 10) + H\} + (10H + 8) + (10H + 6) = 230$
 $\Rightarrow 21H + 104 = 230 \Rightarrow 21H = 126$
 $\Rightarrow H = 6.$

146. Let the missing digit be x . Then, 1 (carried over) + 3 + x + $x = 10 + x \Rightarrow x = 6.$

147. Clearly, $M = 0$ since $304 \times 4 = 1216.$

148. $5p9 + 327 + 2q8 = 1114 \Rightarrow (500 + 10p + 9) + (327) + (200 + 10q + 8) = 1114$

$$\begin{aligned} \Rightarrow 10(p + q) + 1044 &= 1114 \\ \Rightarrow 10(p + q) &= 70 \Rightarrow (p + q) = 7 \\ \Rightarrow \text{Maximum value of } q &\text{ is } 7 \\ [\text{As minimum value of } p &= 0] \end{aligned}$$

149. $5P7 + 8Q9 + R32 = 1928 \Rightarrow (500 + 10P + 7) + (800 + 10Q + 9) + (100R + 30 + 2) = 1928$

$$\begin{aligned} \Rightarrow 10P + 10Q + 100R + 1348 &= 1928 \\ \Rightarrow 10(P + Q + 10R) &= 580 \\ \Rightarrow P + Q + 10R = 58 \Rightarrow R &= 5 \text{ and} \\ P + Q = 8 \text{ or } R = 4 \text{ and } P + Q &= 18 \end{aligned}$$

$$\Rightarrow \text{Maximum value of } Q \text{ is } 9$$

[for $P = 9$ in second case]

150. Let $\frac{1x5y4}{148} = 78.$

$$\begin{aligned} \text{Then, } 10000 + 1000x + 500 + 10y + 4 &= 148 \times 78 \\ \Rightarrow 10000 + 1000x + 500 + 10y + 4 &= 11544 \\ = 10000 + 1000 + 500 + 40 + 4 \\ \Rightarrow 1000x &= 1000 \Rightarrow x = 1. \end{aligned}$$

151. Let $6x43 - 46y9 = 1904.$

Clearly, $y = 3$ and $x = 5.$

Hence * must be replaced by 5.

152. $5P9 - 7Q2 + 9R6 = 823$

$$\begin{aligned} \Rightarrow (500 + 10P + 9) - (700 + 10Q + 2) + (900 + 10R + 6) &= 823 \\ \Rightarrow (500 + 900 - 700) + 10(P + R - Q) + (9 + 6 - 2) &= 823 \\ \Rightarrow 700 + 10(P + R - Q) = 810 = 700 + 110 \\ \Rightarrow 10(P + R - Q) &= 110 \\ \Rightarrow P + R - Q &= 11 \\ \Rightarrow Q &= (P + R - 11). \end{aligned}$$

To get maximum value of Q we take $P = 9$ and $R = 9.$

This gives $Q = (9 + 9 - 11) = 7.$

Hence, the maximum value of Q is 7.

153. Let the required digit be x . Then

$$\begin{aligned} x + 1x + 2x + x3 + x1 &= 21x \\ \Rightarrow x + 10 + x + 20 + x + 10x + 3 + 10x + 1 &= 200 + 10 + x \\ \Rightarrow 22x = 210 - 34 = 176 \Rightarrow x &= 8. \end{aligned}$$

Hence, the required digit is 8.

154. It is given that $D = 0$. So, we have $A = 5.$

So, 1 is carried over.

$$1 + B + C = 10 + C \Rightarrow 1 + B = 10 \Rightarrow B = 9.$$

Now, $1 + C + C = A \Rightarrow 1 + 2C = 5 \Rightarrow 2C = 4 \Rightarrow C = 2.$

Hence $B = 9.$

$$\begin{array}{r} \textcircled{1} \textcircled{1} \\ CBA = 5 \\ + CCA = 5 \\ \hline AC0 \end{array}$$

155. We have, $C = 2.$

156. Since 13b7 is divisible by 11, we have

$$(7 + 3) - (b + 1) = 0 \Rightarrow 9 - b = 0 \Rightarrow b = 9.$$

Putting $b = 9$, $a + 8 = 9$ we get $a = 1$. Hence, $(a + b) = (1 + 9) = 10.$

$$\begin{array}{r} 4a3 \\ 984 \\ \hline 13b7 \end{array}$$

157. Clearly, we have $ab \times b = 24$ and $ab \times a = 12.$

$$\therefore \frac{ab \times b}{ab \times a} = \frac{24}{12} \Rightarrow \frac{b}{a} = \frac{2}{1} \Rightarrow a = 1, b = 2.$$

158. Clearly, $111 \times 1 = 111 \neq 8111.$

But, $999 \times 9 = 8991$. Hence, we have 9 in place of *.

159. $(1 * 2) = 1 + 6 \times 2 = 1 + 12 = 13.$

$$(1 * 2) * 3 = 13 * 3 = 13 + 6 \times 3 = 13 + 18 = 31.$$

160. $|\underline{8} - \underline{7} - \underline{6}| = 8 \times 7 \times |\underline{6} - 7 \times \underline{6} - \underline{6}|$

$$= (56 - 7 - 1) \times |\underline{6}| = 48 \times |\underline{6}| = 6 \times 8 \times |\underline{6}|.$$

161. Highest power of 3 in $99! = \left[\frac{99}{3} \right] + \left[\frac{99}{3^2} \right] + \left[\frac{99}{3^3} \right] + \left[\frac{99}{3^4} \right]$
 $= \left[\frac{99}{3} \right] + \left[\frac{99}{9} \right] + \left[\frac{99}{27} \right] + \left[\frac{99}{81} \right]$
 $= 33 + 11 + 3 + 1 = 48.$

Since $9 = 3^2$, so highest power of 9 dividing $99! = \frac{48}{2} = 24.$

162. Every number from $\underline{5}$ onwards is completely divisible by 5.

$\therefore (\underline{5} + \underline{6} + \underline{7} + \dots + \underline{100})$ is completely divisible by 5.

And,

$$(\underline{1} + \underline{2} + \underline{3} + \underline{4}) = (1 + 2 + 3 \times 2 \times 1 + 4 \times 3 \times 2 \times 1) = (1 + 2 + 6 + 24) = 33.$$

Clearly, 33 when divided by 5 leaves a remainder 3.

Hence, $(\underline{1} + \underline{2} + \underline{3} + \underline{4} + \underline{5} + \dots + \underline{100})$ when divided by 5 leaves a remainder 3.

163. $6^{10} \times 7^{17} \times 11^{27} = (2 \times 3)^{10} \times 7^{17} \times 11^{27} = 2^{10} \times 3^{10} \times 7^{17} \times 11^{27}.$
 Number of prime factors in the given expression = $(10 + 10 + 17 + 27) = 64.$

164. $(30)^7 \times (22)^5 \times (34)^{11} = (2 \times 3 \times 5)^7 \times (2 \times 11)^5 \times (2 \times 17)^{11}$
 $= 2^{(7+5+11)} \times 3^7 \times 5^7 \times 11^5 \times 17^{11}$
 $= (2^{23} \times 3^7 \times 5^7 \times 11^5 \times 17^{11}).$

Number of prime factors = $(23 + 7 + 7 + 5 + 11) = 53.$

165. Let $x \times 48 = 173 \times 240$. Then, $x = \frac{173 \times 240}{48} = 173 \times 5 = 865.$

166. $(1000 + x) > (1000 \times x)$. Clearly, $x = 1.$

167. Let the original number be x . Then,

$$\frac{(x+7) \times 5}{9} - 3 = 12 \Rightarrow \frac{5x+35}{9} = 15 \Rightarrow 5x+35 = 135$$

$$\Rightarrow 5x = 100 \Rightarrow x = 20.$$

168. 38950 is completely divisible by 410 only.

Hence ₹ 410 is the correct answer.

169. Let the four consecutive even numbers be $a, a + 2, a + 4$ and $a + 6.$

$$\text{Then, } a + a + 2 + a + 4 + a + 6 = 180 \Rightarrow 4a = 168$$

$$\Rightarrow a = 42.$$

So, these numbers are 42, 44, 46 and 48.

Sum of next four consecutive even numbers = $(50 + 52 + 54 + 56) = 212.$

170. Number on thumbs = 1, 9, 17, 25,

This is an AP in which $a = 1$ and $d = (9 - 1) = 8.$